Tree Builder: Answer Key

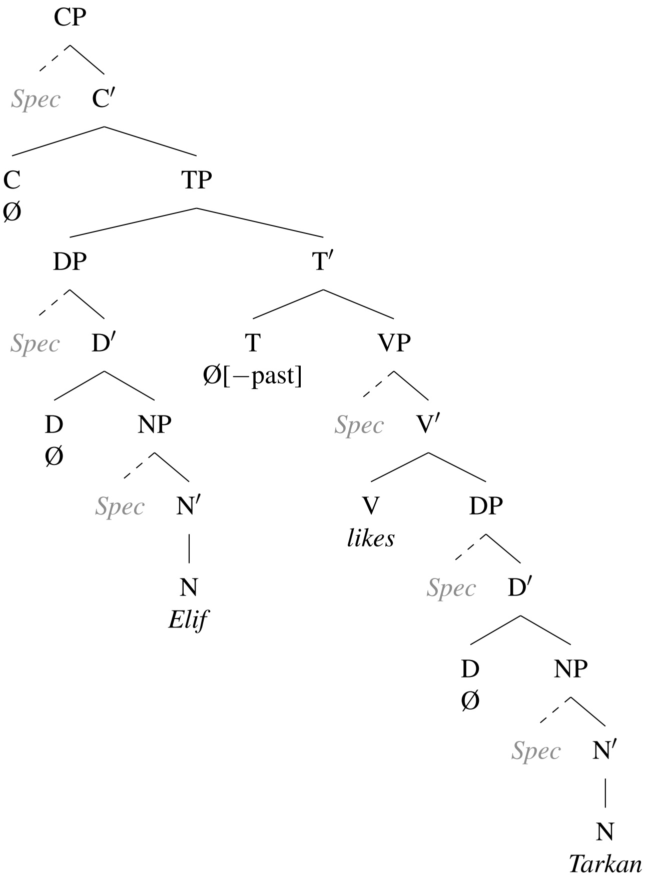
Reading: Andrew Carnie (2013), *Syntax: A Generative Introduction*, 3rd ed., Chapters 6 and 7

Conventions. Every clause is a CP; the main-clause C is Ø. The subject sits in Spec,TP. Modals sit in T, and so does *to*[−tense] in a non-finite clause; with no modal or auxiliary, T is Ø with a tense feature, Ø[+past] or Ø[−past], and the verb keeps the tense ending. D heads DP; names and bare nouns have an empty D (Ø). Complements are sisters of the head; each adjunct gets its own X′. Empty specifier positions are drawn as dashed *Spec* branches. No movement.

1. Level 1: the clause spine

1.1 Elif likes Tarkan.

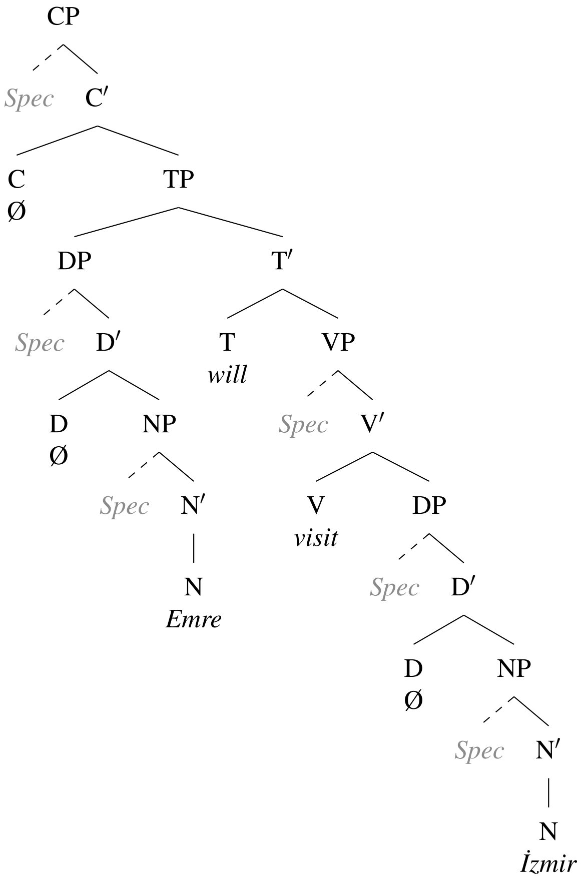
Practice: Ø C, Ø T, names with Ø D as subject and object


CP
Spec
C′
C
Ø
TP
DP
T′
Spec
D′
T
Ø[−past]
VP
D
Ø
NP
Spec
V′
Spec
N′
V
likes
DP
N
Elif
Spec
D′
D
Ø
NP
Spec
N′
N
Tarkan

[CP [C′ [C Ø] [TP [DP [D′ [D Ø] [NP [N′ [N *Elif*]]]]] [T′ [T Ø[−past]] [VP [V′ [V *likes*] [DP [D′ [D Ø] [NP [N′ [N *Tarkan*]]]]]]]]]]]

1.2 Emre will visit İzmir.

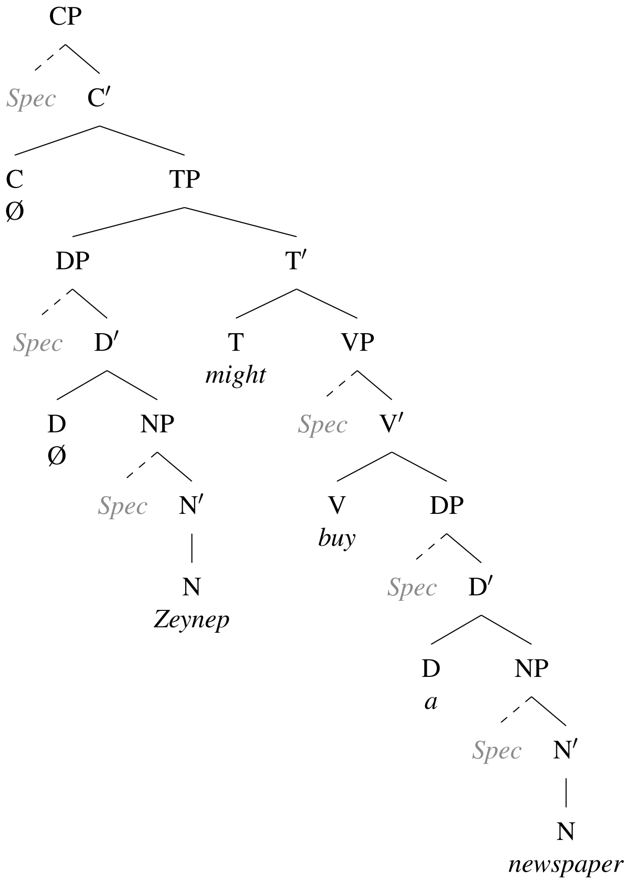
Practice: modal will in T, place name as object DP


CP
Spec
C′
C
Ø
TP
DP
T′
Spec
D′
T
will
VP
D
Ø
NP
Spec
V′
Spec
N′
V
visit
DP
N
Emre
Spec
D′
D
Ø
NP
Spec
N′
N
İzmir

[$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ Ø] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Emre*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ *will*] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *visit*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *İzmir*]]]]]]]]]]]

1.3 Zeynep might buy a newspaper.

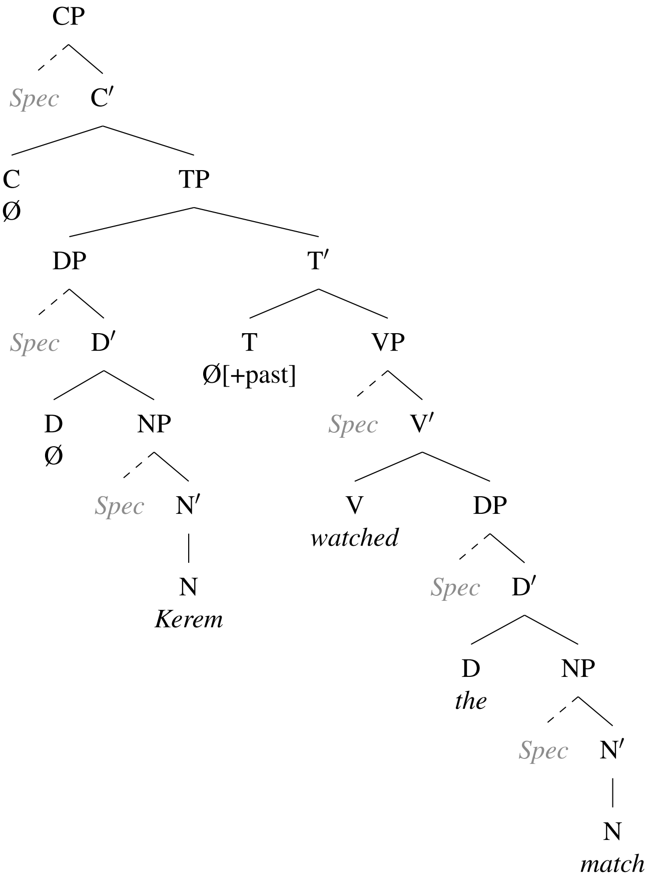
Practice: modal might in T, object DP with overt D a


CP
Spec
C′
C
Ø
TP
DP
Spec
D′
D
Ø
NP
Spec
N′
N
Zeynep
T′
T
might
VP
Spec
V′
V
buy
DP
Spec
D′
D
a
NP
Spec
N′
N
newspaper

[$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ Ø] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Zeynep*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ *might*] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *buy*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ *a*] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *newspaper*]]]]]]]]]]]

1.4 Kerem watched the match.

Practice: Ø T with a past-tense verb (no affix lowering), object DP with overt D the

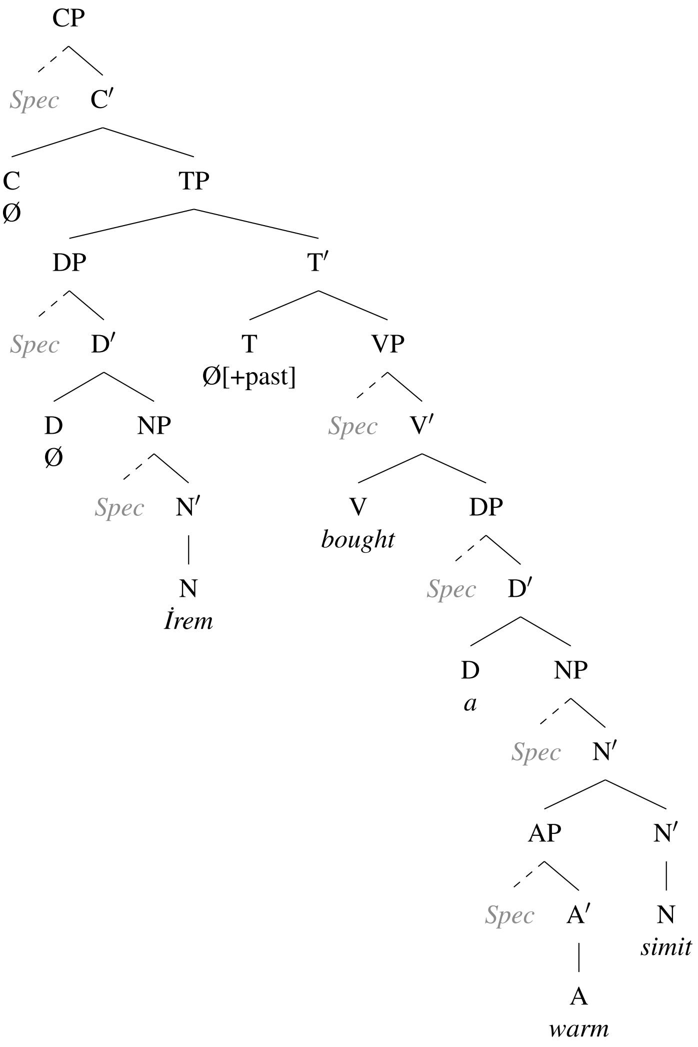

CP
Spec
C′
C
Ø
TP
DP
T′
Spec
D′
T
Ø[+past]
VP
D
Ø
NP
Spec
V′
Spec
N′
V
watched
DP
N
Kerem
Spec
D′
D
the
NP
Spec
N′
N
match

[$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ Ø] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Kerem*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ Ø[+past]] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *watched*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ *the*] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *match*]]]]]]]]]]]

2. Level 2: inside phrases

2.1 İrem bought a warm simit.

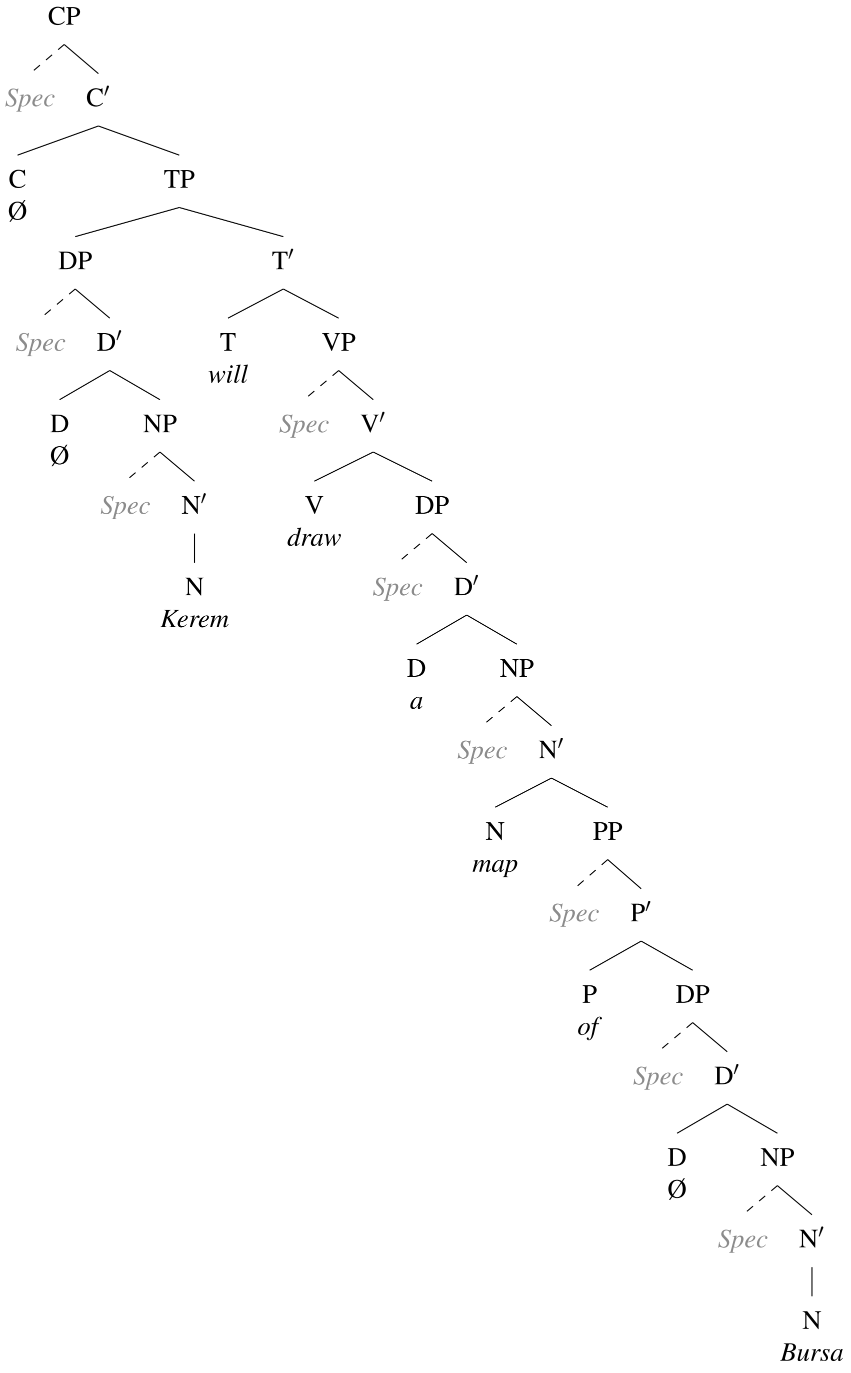
Practice: AP adjunct before a noun: extra N′ above N′–N


CP
Spec
C′
C
Ø
TP
DP
T′
Spec
D′
T
Ø[+past]
VP
D
Ø
NP
Spec
V′
Spec
N′
V
bought
DP
N
İrem
Spec
D′
D
a
NP
Spec
N′
AP
N′
Spec
A′
N
simit
A
warm

[CP [C′ [C Ø] [TP [DP [D′ [D Ø] [NP [N′ [N *İrem*]]]]] [T′ [T Ø[+past]] [VP [V′ [V *bought*] [DP [D′ [D *a*] [NP [N′ [AP [A′ [A *warm*]]] [N′ [N *simit*]]]]]]]]]]]

2.2 Kerem will draw a map of Bursa.

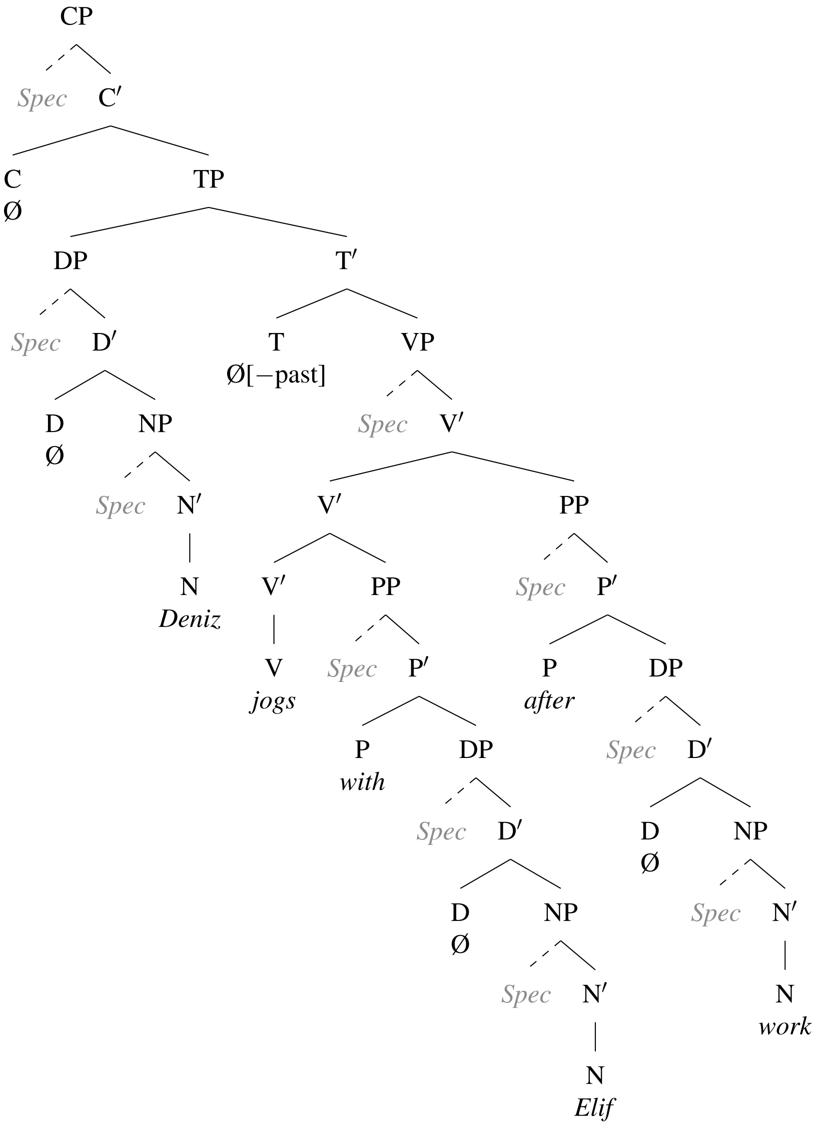
Practice: of-PP as complement of N (sister of N), modal in T


CP
Spec
C′
C
Ø
TP
DP
T′
Spec
D′
T
will
VP
D
Ø
NP
Spec
V′
Spec
N′
V
draw
DP
N
Kerem
Spec
D′
D
a
NP
Spec
N′
N
map
PP
Spec
P′
P
of
DP
Spec
D′
D
Ø
NP
Spec
N′
N
Bursa

[$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ Ø] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Kerem*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ *will*] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *draw*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ *a*] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *map*] [$_{\text{PP}}$ [$_{\text{P}'}$ [$_{\text{P}}$ *of*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Bursa*]]]]]]]]]]]]]]]]]

2.3 *Deniz jogs with Elif after work.*

Practice: two PP adjuncts stacked on V′, same side; the first PP attaches lower

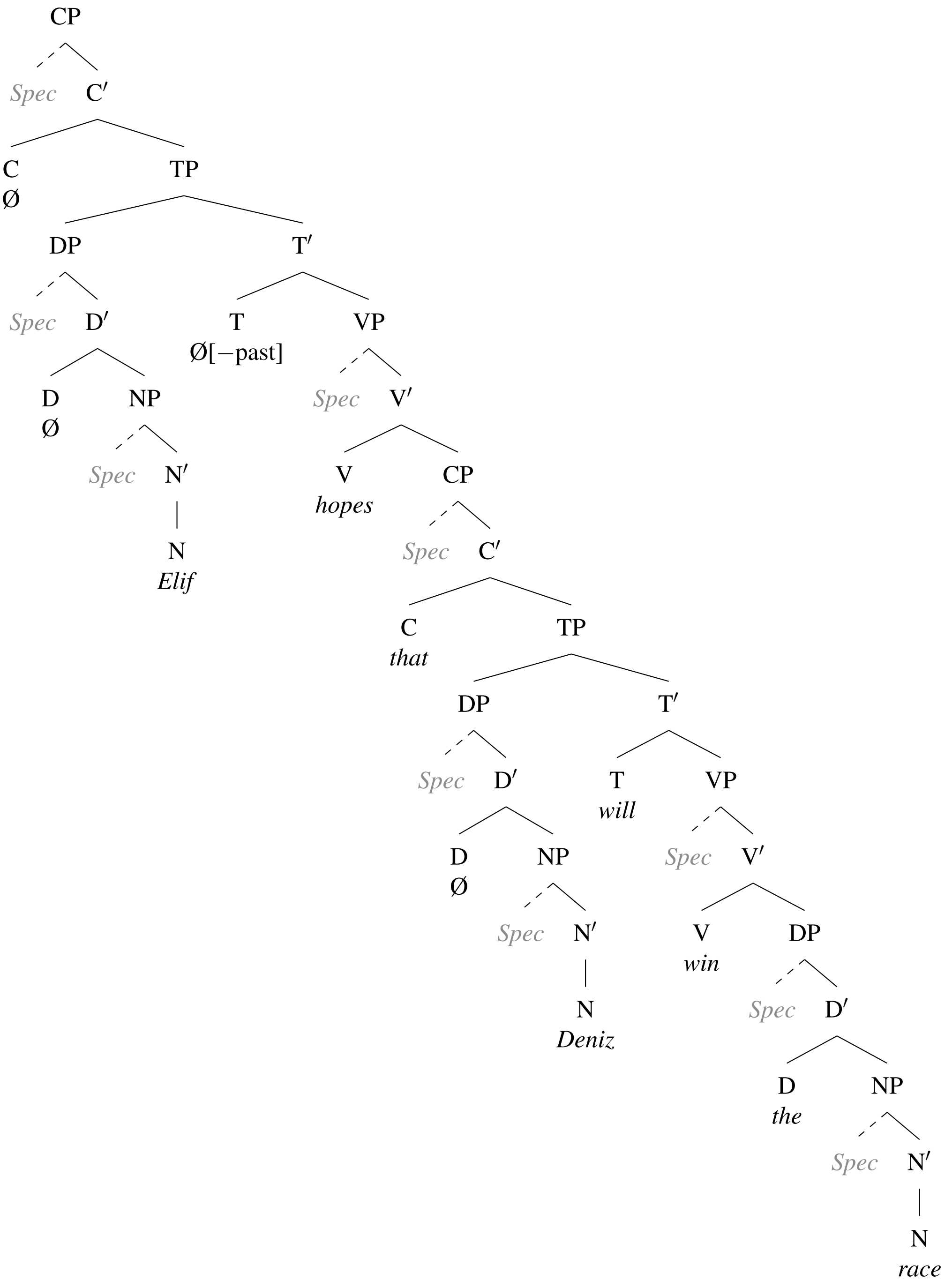

CP
Spec
C′
C
Ø
TP
DP
T′
Spec
D′
T
Ø[−past]
VP
D
Ø
NP
Spec
V′
Spec
N′
V′
PP
N
Deniz
V′
PP
Spec
P′
V
jogs
Spec
P′
P
after
DP
P
with
DP
Spec
D′
Spec
D′
D
Ø
NP
D
Ø
NP
Spec
N′
Spec
N′
N
work
N
Elif

[$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ Ø] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Deniz*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ Ø[−past]] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}'}$ [$_{\text{V}'}$ [$_{\text{V}}$ *jogs*]] [$_{\text{PP}}$ [$_{\text{P}'}$ [$_{\text{P}}$ *with*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Elif*]]]]]]]] [$_{\text{PP}}$ [$_{\text{P}'}$ [$_{\text{P}}$ *after*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *work*]]]]]]]]]]]]]

3. Level 3: embedded clauses

3.1 Elif hopes that Deniz will win the race.

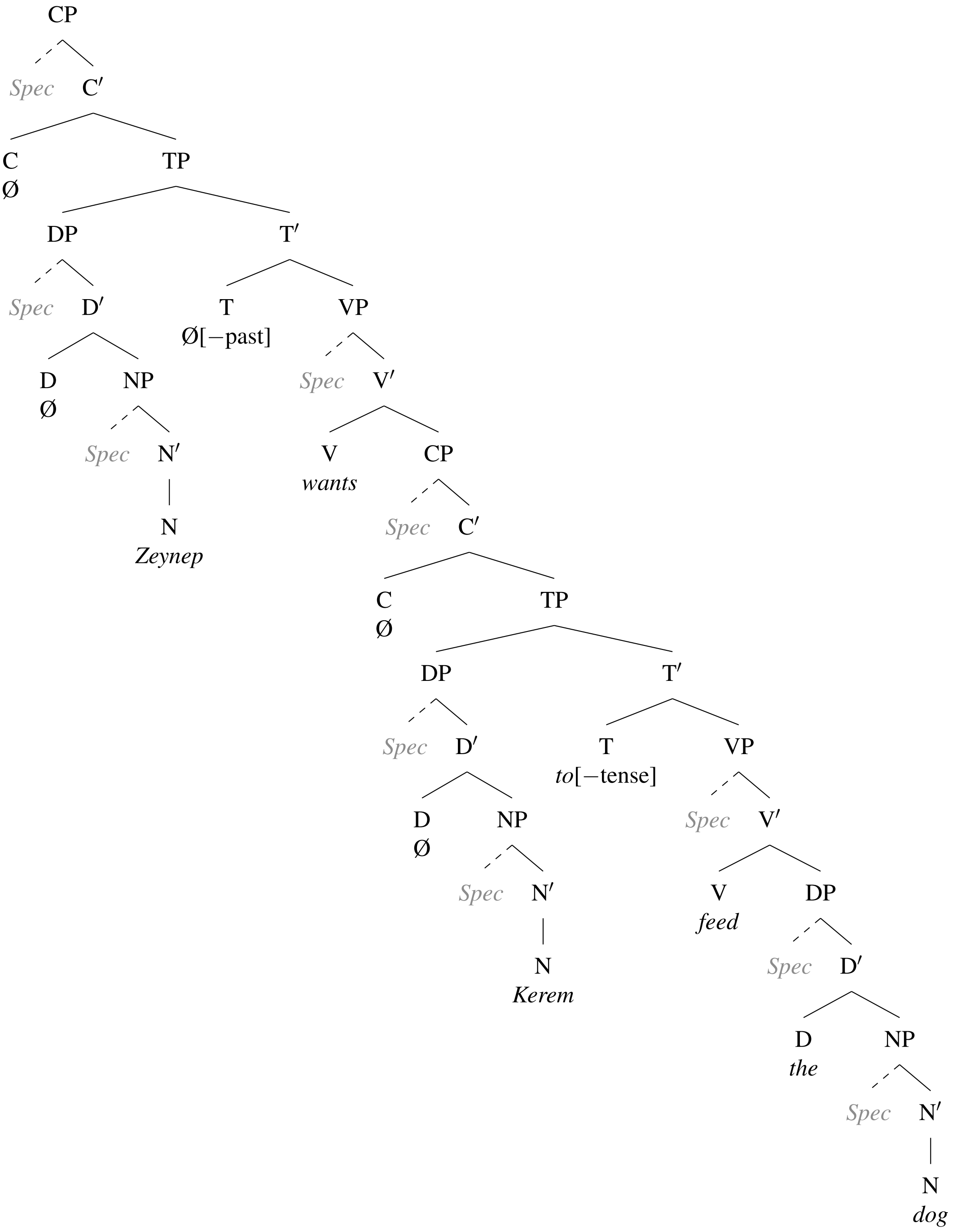
Practice: that-CP as complement of V; modal in the embedded T


CP
Spec
C′
C
Ø
TP
DP
T′
Spec
D′
T
Ø[−past]
VP
D
Ø
NP
Spec
V′
Spec
N′
V
hopes
CP
N
Elif
Spec
C′
C
that
TP
DP
T′
Spec
D′
T
will
VP
D
Ø
NP
Spec
V′
Spec
N′
V
win
DP
N
Deniz
Spec
D′
D
the
NP
Spec
N′
N
race

[$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ Ø] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Elif*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ Ø[−past]] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *hopes*] [$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ *that*] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Deniz*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ *will*] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *win*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ *the*] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *race*]]]]]]]]]]]]]]]]]

3.2 Zeynep wants Kerem to feed the dog.

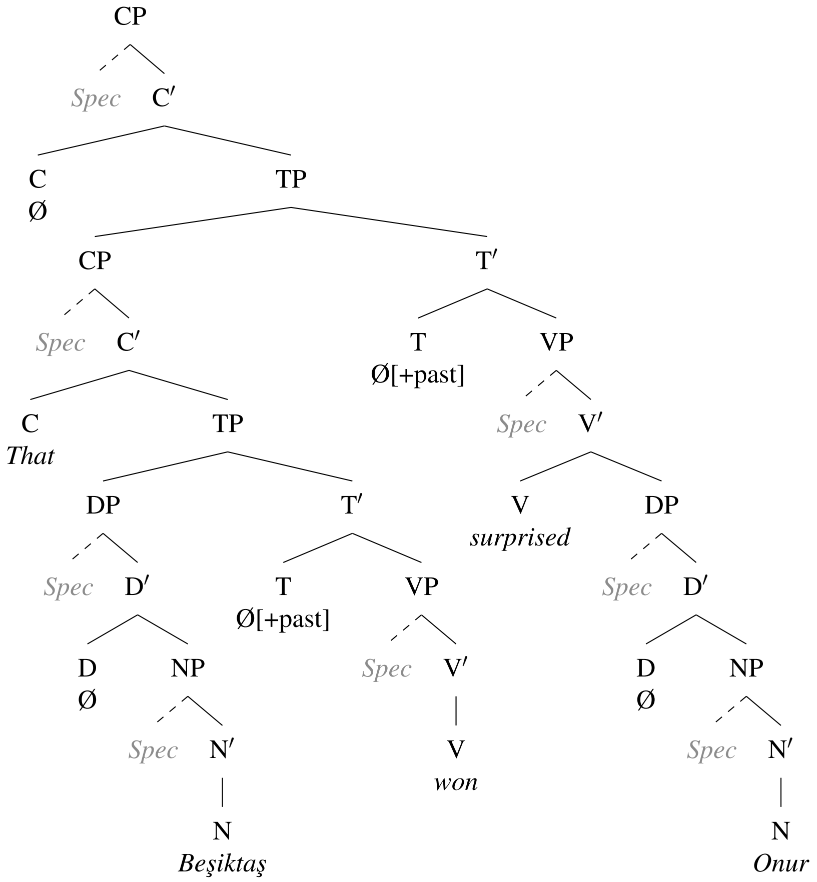
Practice: non-finite CP complement: to in T, the embedded subject in Spec,TP, an empty embedded C


CP
Spec
C′
C
Ø
TP
DP
T′
Spec
D′
T
Ø[−past]
VP
D
Ø
NP
Spec
V′
Spec
N′
V
wants
CP
N
Zeynep
Spec
C′
C
Ø
TP
DP
T′
Spec
D′
T
to[−tense]
VP
D
Ø
NP
Spec
V′
Spec
N′
V
feed
DP
N
Kerem
Spec
D′
D
the
NP
Spec
N′
N
dog

$[_{\text{CP}}$ $[_{\text{C}'}$ $[_{\text{C}}$ Ø] $[_{\text{TP}}$ $[_{\text{DP}}$ $[_{\text{D}'}$ $[_{\text{D}}$ Ø] $[_{\text{NP}}$ $[_{\text{N}'}$ $[_{\text{N}}$ *Zeynep*]]]]] $[_{\text{T}'}$ $[_{\text{T}}$ Ø[−past]] $[_{\text{VP}}$ $[_{\text{V}'}$ $[_{\text{V}}$ *wants*] $[_{\text{CP}}$ $[_{\text{C}'}$ $[_{\text{C}}$ Ø] $[_{\text{TP}}$ $[_{\text{DP}}$ $[_{\text{D}'}$ $[_{\text{D}}$ Ø] $[_{\text{NP}}$ $[_{\text{N}'}$ $[_{\text{N}}$ *Kerem*]]]]] $[_{\text{T}'}$ $[_{\text{T}}$ *to*[−tense]] $[_{\text{VP}}$ $[_{\text{V}'}$ $[_{\text{V}}$ *feed*] $[_{\text{DP}}$ $[_{\text{D}'}$ $[_{\text{D}}$ *the*] $[_{\text{NP}}$ $[_{\text{N}'}$ $[_{\text{N}}$ *dog*]]]]]]]]]]]]]]]]]

3.3 That Beşiktaş won surprised Onur.

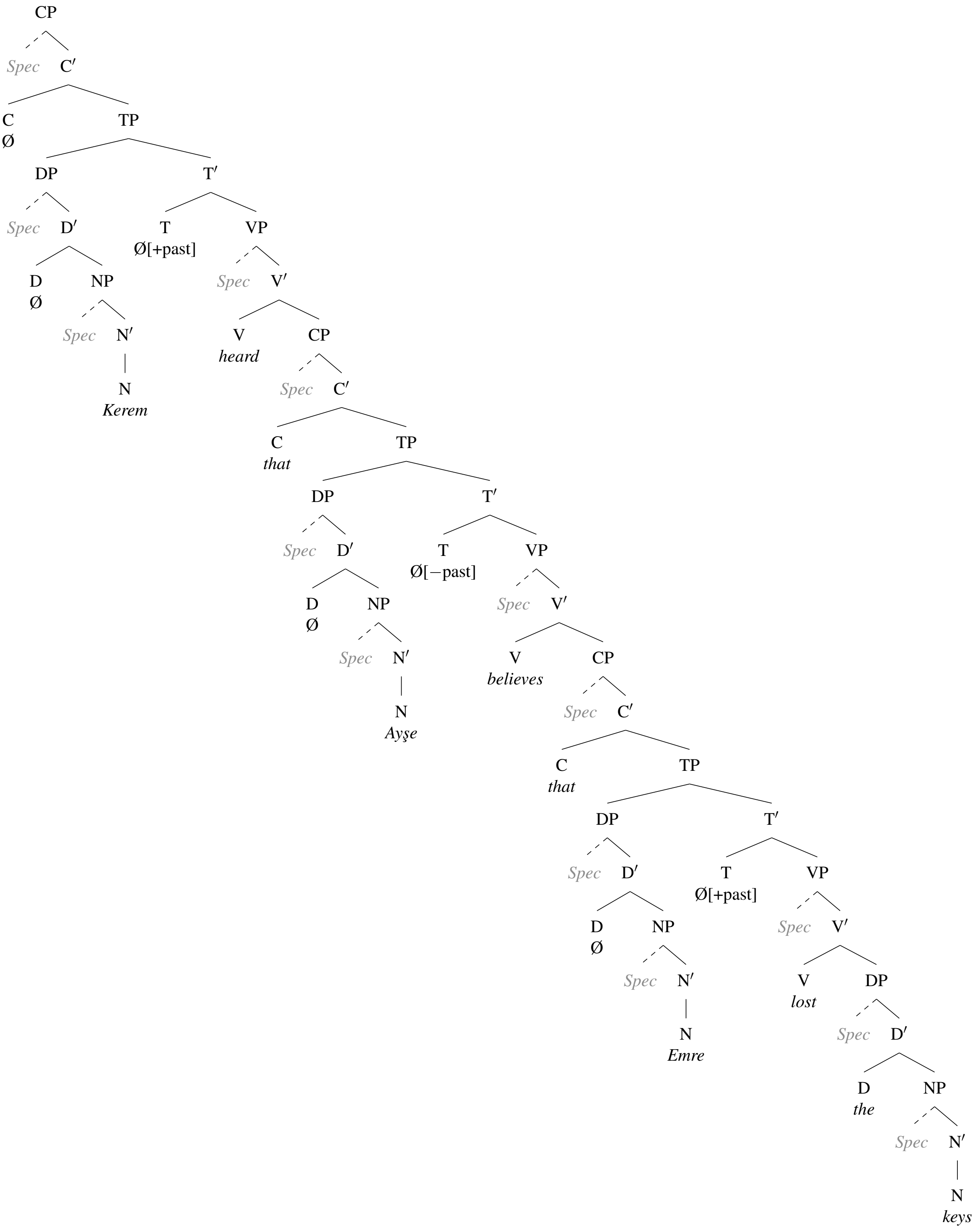
Practice: sentential subject: a that-CP in Spec,TP


CP
Spec
C′
C
Ø
TP
CP
T′
Spec
C′
T
Ø[+past]
VP
C
That
TP
Spec
V′
DP
T′
V
surprised
DP
Spec
D′
T
Ø[+past]
VP
Spec
D′
D
Ø
NP
Spec
V′
D
Ø
NP
Spec
N′
V
won
Spec
N′
N
Beşiktaş
N
Onur

[$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ Ø] [$_{\text{TP}}$ [$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ *That*] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Beşiktaş*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ Ø[+past]] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *won*]]]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ Ø[+past]] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *surprised*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Onur*]]]]]]]]]]

3.4 Kerem heard that Ayşe believes that Emre lost the keys.

Practice: two levels of embedding: a that-CP inside a that-CP


CP
Spec
C′
C
Ø
TP
DP
Spec
D′
D
Ø
NP
Spec
N′
N
Kerem
T′
T
Ø[+past]
VP
Spec
V′
V
heard
CP
Spec
C′
C
that
TP
DP
Spec
D′
D
Ø
NP
Spec
N′
N
Ayşe
T′
T
Ø[−past]
VP
Spec
V′
V
believes
CP
Spec
C′
C
that
TP
DP
Spec
D′
D
Ø
NP
Spec
N′
N
Emre
T′
T
Ø[+past]
VP
Spec
V′
V
lost
DP
Spec
D′
D
the
NP
Spec
N′
N
keys

[$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ Ø] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Kerem*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ Ø[+past]] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *heard*] [$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ *that*] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Ayşe*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ Ø[−past]] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *believes*] [$_{\text{CP}}$ [$_{\text{C}'}$ [$_{\text{C}}$ *that*] [$_{\text{TP}}$ [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ Ø] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *Emre*]]]]] [$_{\text{T}'}$ [$_{\text{T}}$ Ø[+past]] [$_{\text{VP}}$ [$_{\text{V}'}$ [$_{\text{V}}$ *lost*] [$_{\text{DP}}$ [$_{\text{D}'}$ [$_{\text{D}}$ *the*] [$_{\text{NP}}$ [$_{\text{N}'}$ [$_{\text{N}}$ *keys*]]]]]]]]]]]]]]]]]]]]]]]